



Analog HD Camera

User's Manual

Foreword

General

This manual introduces the functions and operations of the Analog HD camera (hereinafter referred to as "the device").

Safety Instructions

The following categorized signal words with defined meaning might appear in the manual.

Signal Words	Meaning
 DANGER	Indicates a high potential hazard that, if not avoided, will result in death or serious injury.
 WARNING	Indicates a medium or low potential hazard which, if not avoided, could result in slight or moderate injury.
 CAUTION	Indicates a potential risk that, if not avoided, could result in property damage, data loss, lower performance, or unpredictable result.
 TIPS	Provides methods to help you solve a problem or save you time.
 NOTE	Provides additional information as the emphasis and supplement to the text.

About the Manual

- The manual is for reference only. If there is an inconsistency between the manual and the actual product, the actual product shall prevail.
- We are not liable for any loss caused by the operations that do not comply with the manual.
- The manual would be updated according to the latest laws and regulations of related jurisdictions. For detailed information, refer to the paper manual, CD-ROM, QR code or our official website. If there is an inconsistency between the paper manual and the electronic version, the electronic version shall prevail.
- All the designs and software are subject to change without prior written notice. The product updates might cause some differences between the actual product and the manual. Please contact customer service for the latest program and supplementary documentation.
- There still might be deviations in technical data, functions, and operations description, or errors in print. If there is any doubt or dispute, we reserve the right of a final explanation.

- Upgrade the reader software or try other mainstream reader software if the manual (in PDF format) cannot be opened.
- All trademarks, registered trademarks and the company names in the manual are the properties of their respective owners.
- Please visit our website, and contact the supplier or customer service if there is any problem occurring when using the device.
- If there is any uncertainty or controversy, we reserve the right of a final explanation.

Important Safeguards and Warnings



Electrical Safety

- All installation and operation should conform to your local electrical safety codes.
- The power source shall conform to the requirement of the Safety Extra Low Voltage (SELV) standard, and supply power with rated voltage which conforms to Limited power Source requirement according to ICE62368-1. Note that the power supply requirement is subject to the device label.
- A readily accessible disconnect device shall be incorporated in the building installation wiring.
- Make sure that the power adapter meets the device operating voltage requirement before powering up the device (The material and length of the power cable might influence the device voltage).
- Prevent the power cable from being trampled or pressed, especially the plug, power socket and the junction extruded from the device.
- We assume no liability or responsibility for all the fires or electrical shock caused by improper handling or installation.

Operating Requirements

- Do not aim the device at strong light to focus, such as lamp light and sun light.
- Transport, use and store the device within the range of allowed humidity and temperature.
- Keep the device away from water or other liquid to avoid damages to the internal components.
- Keep sound ventilation to avoid heat accumulation.
- Heavy stress, violent vibration or water splash are not allowed during transportation, storage and installation.
- Pack the device with standard factory packaging or the equivalent material when transporting the device.
- You are recommended to use the device together with lightning arrester to improve lightning protection effect.
- You are recommended to ground the device to enhance reliability.
- You are recommended to use qualified video transmission cable to improve video quality, and use RG59 coaxial cable or higher standard.



WARNING

- Use standard components or accessories provided by the manufacturer and make sure that the device is installed and maintained by professional engineers.
- The surface of the image sensor should not be exposed to laser beam radiation in an environment where a laser beam device is used.

- Do not provide two or more power supply sources for the device; otherwise it might damage the device.
- If PoC power supply is used, do not connect any other device between the device and PoC transceiver including UTC, Balun, optical transceiver, distributor and convertor and so on; otherwise, the device might get burned.
- PoC supply voltage is up to 52V. Do not dismantle the device during normal operation; otherwise it might cause danger to both device and users due to high voltage.

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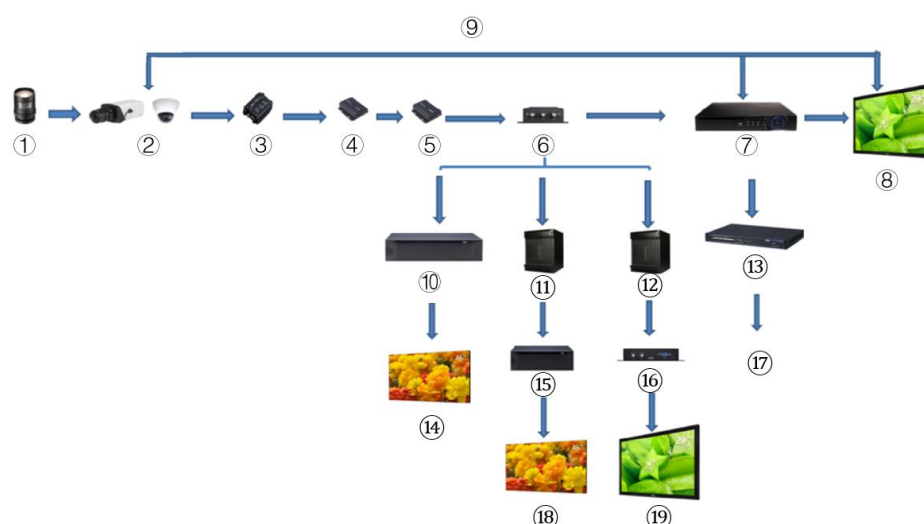
1 Overview

1.1 Introduction

The devices comply with the HDCVI standard and support the transmission of video and control signal over coaxial cable. The devices produce video signal with megapixel resolution and require connected DVRs to achieve high-speed, long-distance, and zero-lag transmission of the signal. They are applicable to various scenes, such as roads, warehouses, underground parking lots, bars, pipelines, and gas stations.

1.2 Application

Application scenario



Application scenario

No.	Name	No.	Name	No.	Name
1	(Optional) Lens	8	Display Screen	15	Splicer
2	HDCVI Products	9	Direct Connection	16	Convertor
3	(Optional) Surge Protection Device	10	Integrated Video Platform	17	Ethernet
4	(Optional) Optical Transceiver (Send)	11	Matrix	18	Splicing Screen
5	(Optional) Optical Transceiver (Receive)	12	Matrix	19	Display Screen
6	(Optional) Distributor	13	Switch	—	—
7	HCVR Products	14	Splicing Screen	—	—

1.3 Transmission Distance

Transmission distance

Cable		720P	1080P	4MP/4K
Coaxial Cable	RG6 (75-5)	1200 m	800 m	700 m
	RG59 (75-3)	800 m	500 m	500 m
UTP	CAT6	450 m	300 m	300 m

Transmission distance of PoC HDCVI powered by PoC DVR

PoC DVR Series	PoC Mode	RG59	RG6
Full series	AT	100 m	100 m
	AF	200 m	200 m

2 Cable Connection



Cable types might vary with different cameras, and the actual product shall prevail.

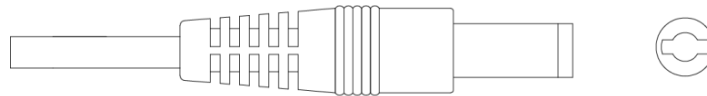
2.1 Power Output

Supplies 12V DC power.



- Ensure that power consumption of devices connected to this port is below 2W.
- Ensure that supply frequency of devices connected to this port is higher than 1MHz, such as sound pick-up, temperature/humidity sensor and other devices without power consumption change. It might cause image flickering if this port is connected to devices with supply frequency less than 1MHz, such as fan, Hall sensor, loudspeaker, motor and other electromechanical devices with power consumption change.

Power output



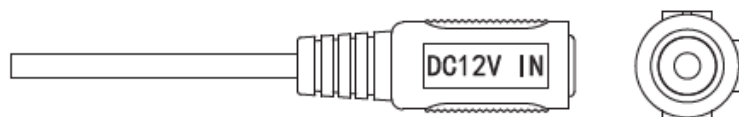
2.2 12V DC Power Input Port

Inputs 12V DC power.



Device abnormality or damage could occur if power is not supplied correctly for 12V DC power input port. Be sure to supply power as instructed in the manual.

12V DC power input port



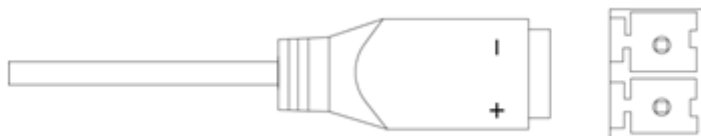
2.3 24V AC Power Input Port

Inputs 24V AC power.



Device abnormality or damage could occur if power is not supplied correctly. Please be sure to supply power as instructed in the manual.

24V AC power input port



2.4 Video Output Port

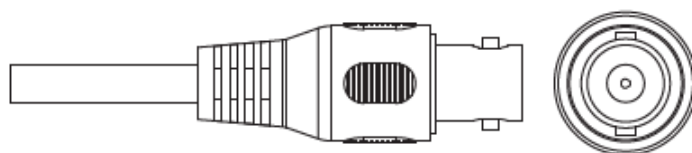
Connects to the DVR to output video signal.



WARNING

- When the device is in the condition of PoC power supply, do not connect any other device between the device and PoC DVR or PoC transceiver including UTC, Balun, optical transceiver, distributor and convertor and so on; otherwise, the device might get burned.
- PoC power supply is with high voltage. Do not dismantle the device during normal operation; otherwise it might cause danger to both device and users due to high voltage.

Video output port



2.5 Audio Input Port

Connects to sound pickup devices to receive analog audio signal.

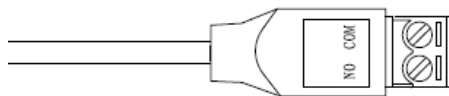
Audio input port



2.6 Alarm Output Port

Connects to external alarm devices such as siren to trigger alarms.

Alarm output port



3 General Configuration and Operation

Power up the device and connect it to the DVR with coaxial cable, and then the live interface is displayed. Then you can start configuring HDCVI cameras on the DVR.



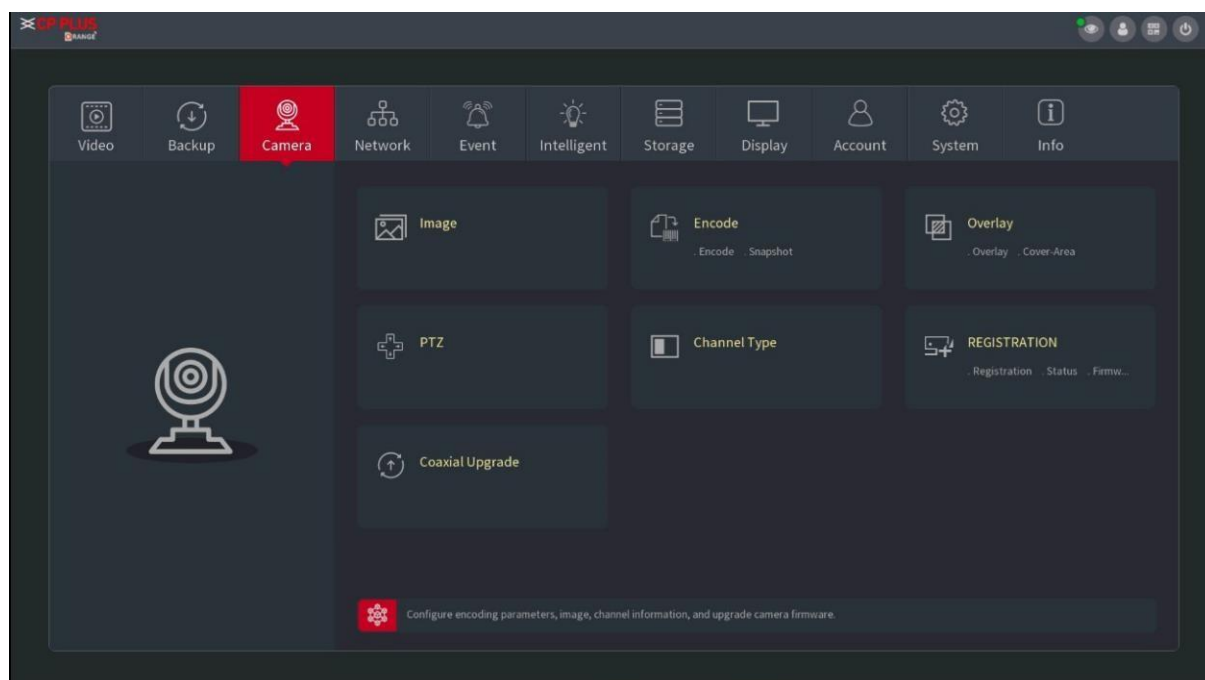
- The No. of the coaxial ports on DVR will display at the lower-left corner of each window to indicate the corresponding camera.
- Ports might vary depending on the DVR models, and the actual product shall prevail.

3.1 Entering DVR Main Menu

Right-click on the live interface, and the shortcut menu is displayed.

Click **Main Menu** and then log in to the system. The main menu of DVR is displayed.

DVR main menu



3.2 Setting Audio Input



Audio input is available on select models.

On the **Main Menu** interface, select **CAMERA > ENCODE > Encode**.

On the **Channel I** drop-down list, select the device that you want to configure according to the coaxial port No.

Under Main Stream, click More Setting.

Encode setting

On the **More Setting** interface, enable **Audio Encode** function and then configure the audio settings. In the **Audio Format** list, leave it as default; in the **Audio Source** list, select **HDCVI**. Click **Save**.

More setting

On the **Encode** interface, click **Apply**.

3.3 Operating PTZ Control Panel

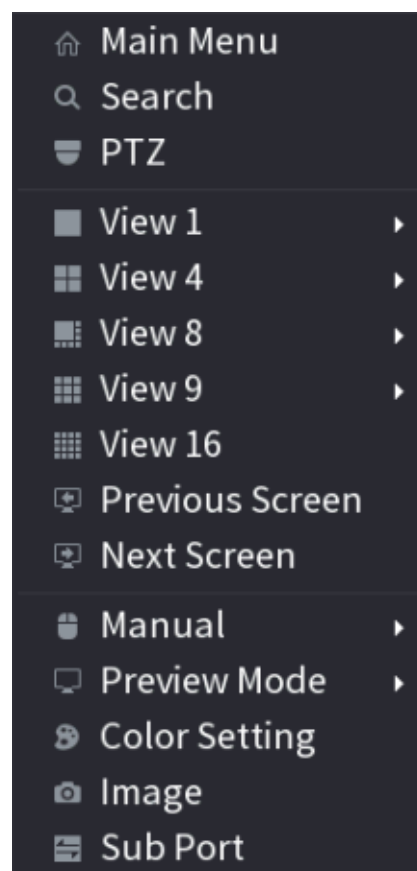
3.3.1 Operating OSD Menu



- The OSD menus of different cameras might vary, and the actual product shall prevail.
- When you use OSD menu to restore the device to default settings, the resolution, mode, frame rate and language of the device will not be restored.

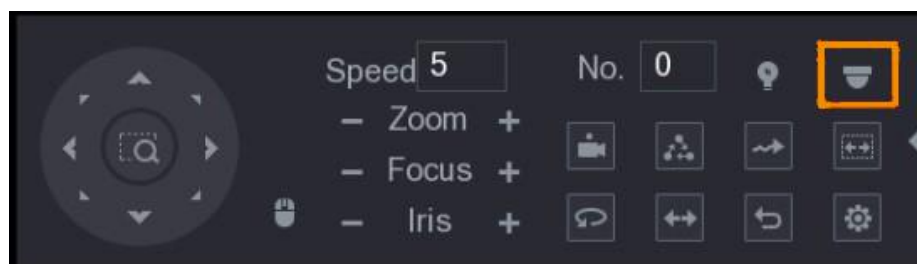
On the live interface, right-click the device that you want to configure. The shortcut menu is displayed.

Shortcut menu



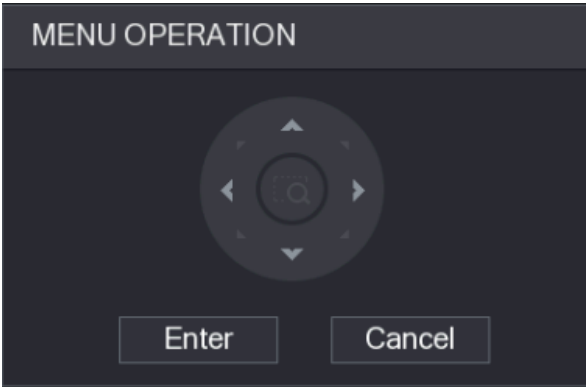
Click **PTZ** and click  to extend the menu.

PTZ setting options



Click . The **MENU OPERATION** panel is displayed.

Menu Operation panel

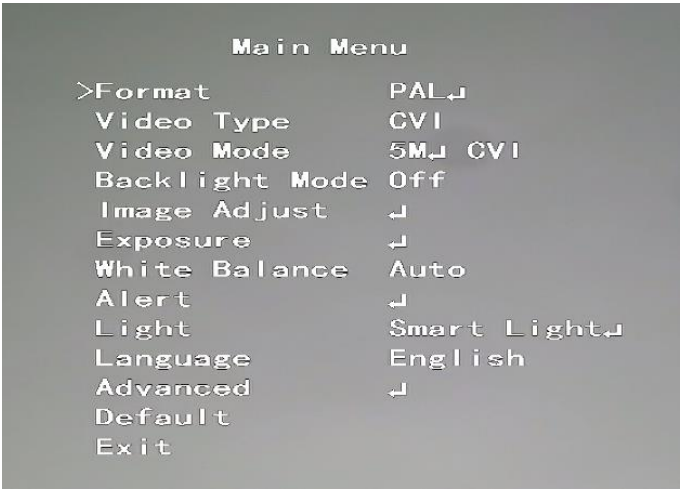


Menu operation panel function

Button	Function	Button	Function
Enter	Enter or confirm an item	 , 	Select item
Cancel	Exit OSD menu	 , 	Change item value

The OSD menu of the corresponding device is displayed on the live interface. If the value of OSD item is "↵", click **Enter** to go to the next level of this item. Click **Return** to go back to the previous level. Click **Cancel** to exit OSD menu without saving the modifications.

OSD menu



3.3.2 Operating Auto Focus (AF)

Parameter of AF

Parameter	Description
Zoom	 : Zoom out.

Parameter	Description
	 : Zoom in.
3 Focus	 : Focus far.
	 : Focus near.
Iris	 : Auto focus.
	 : Open OSD menu.
PTZ movement	Supports eight directions.
	Click  , and then you can control the four directions (left, right, up, and down) of PTZ through mouse operation.
	Click  to unfold PTZ control panel.

4 Full-Color Camera Configuration

This chapter introduces how to configure the working modes of smart light, including auto and manual. Smart light will change the brightness of white light automatically according to the ambient lighting condition to avoid over-exposure. Smart light is only available for full-color camera.

4.1 Enabling/Disabling Smart Light

Smart light is enabled by default. To switch the mode of smart light, enter OSD menu (Figure 3-7) and select **Light > Smart Light**.

4.2 Configuring Smart Light Adjustment

In the smart light mode, configure the maximum brightness level of the smart light, and the device will change brightness automatically according to the ambient lighting condition. You can also configure the sensitivity of the smart light.

4.2.1 Configuring Brightness Level

Step 1 On the OSD menu, select **Light > Smart Light > Level**.

Step 2 Select from **1** to **5** to configure the maximum brightness level.

Step 3 The maximum brightness level is 5 by default.

Click **Return** and then **Exit** to exit the configuration.



You can also configure the brightness level manually in **Light > Manual > Level**.

4.2.2 Configuring Sensitivity

Step 1 Select **Light** > **Smart Light** > **Sensitivity**.

Step 2 Select from **1** to **5** to configure the sensitivity value of the smart light.

Step 3 

- The higher the value is, the easier the smart light will be woken up.
- The sensitivity value is 3 by default.

Click **Return** and then **Exit** to exit the configuration.

5 Smart Dual Illuminators Camera Configuration

Smart dual illuminators camera supports three light modes: **Smart IR&WL**, **WL mode**, **IR mode**. On the OSD menu, select Illuminator, and then view the three different modes.

- **Smart IR&WL**: Turn on the IR light in low ambient brightness. If SMD or IVS has been associated with smart dual illuminators cameras, and when the SMD detects people and vehicles on the screen, or perimeter rules are triggered by the IVS, then the camera will be controlled to turn on the warm light, and the screen automatically switch to the full-color. When SMD or IVS has detected that no rules are triggered by people and vehicles, the camera will turn off the warm light and turn on the IR light to switch to the black & white night vision.
- **WL mode**: If the screen is full-color in low ambient brightness, then the camera is a full-color camera.
- **IR mode**: If the screen is black & white in low ambient brightness, then the camera is an IR camera.

Figure 4-1 Smart IR&WL

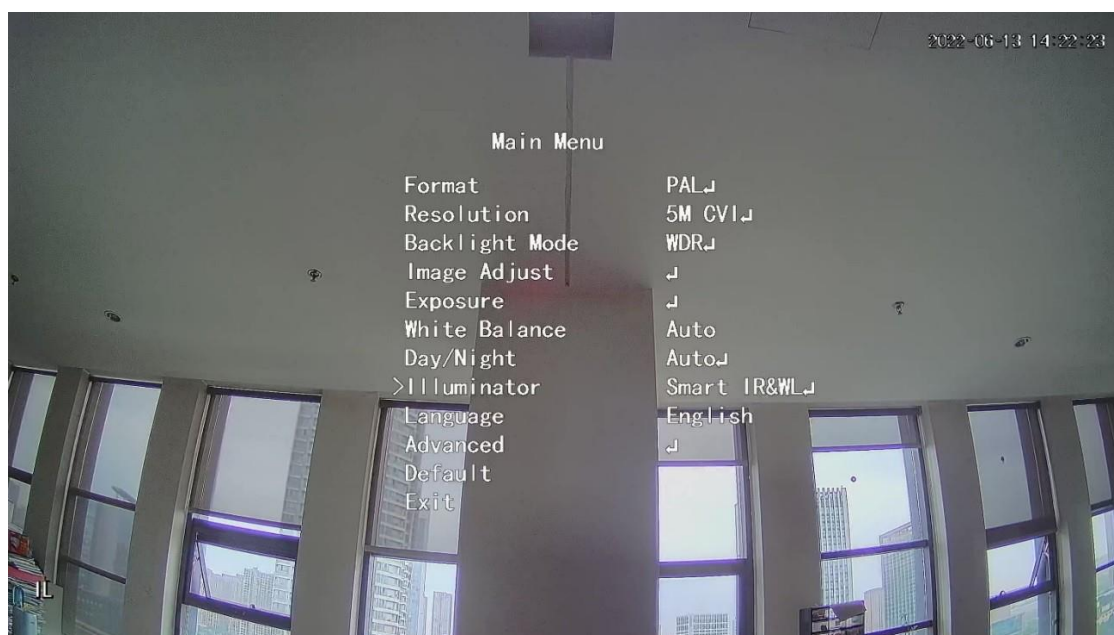


Figure 4-2 WL mode

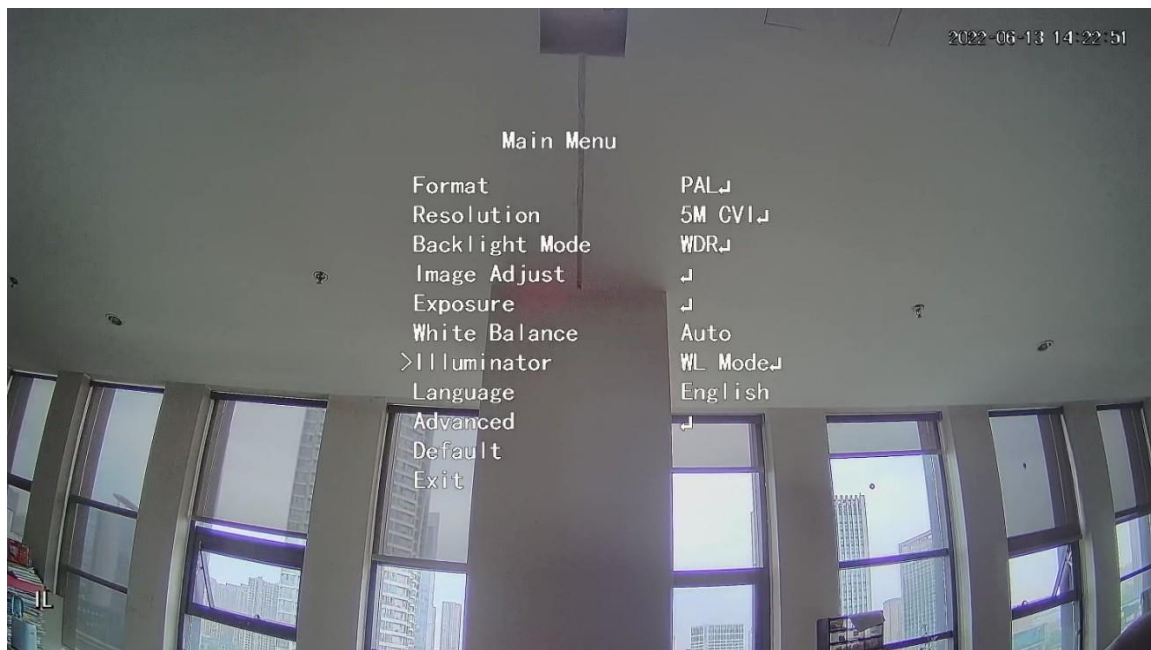
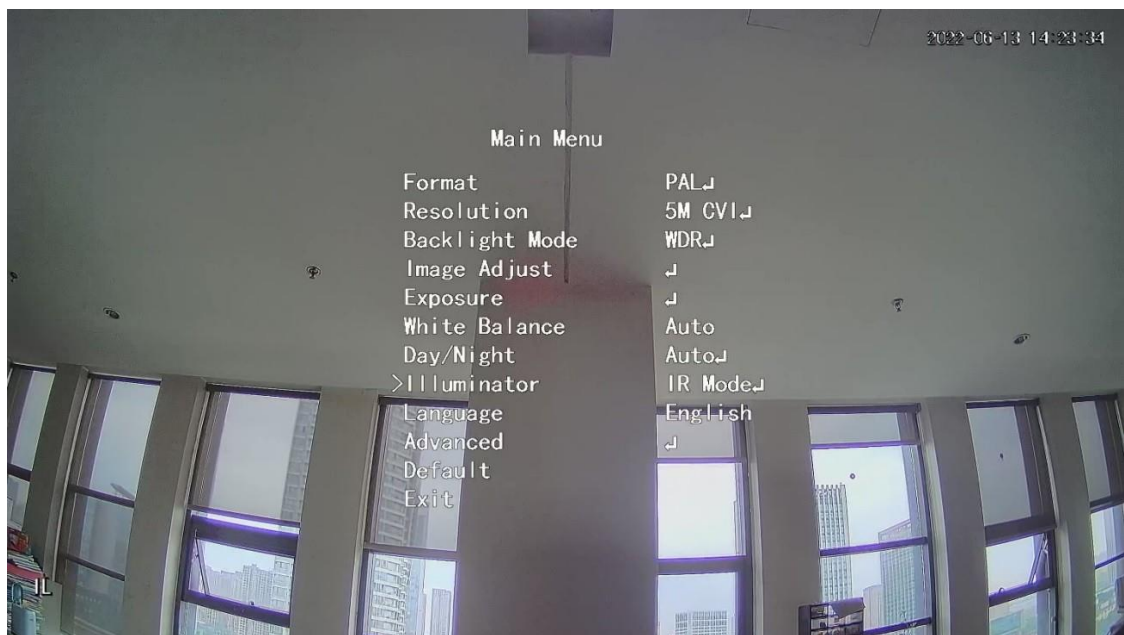


Figure 4-3 IR mode

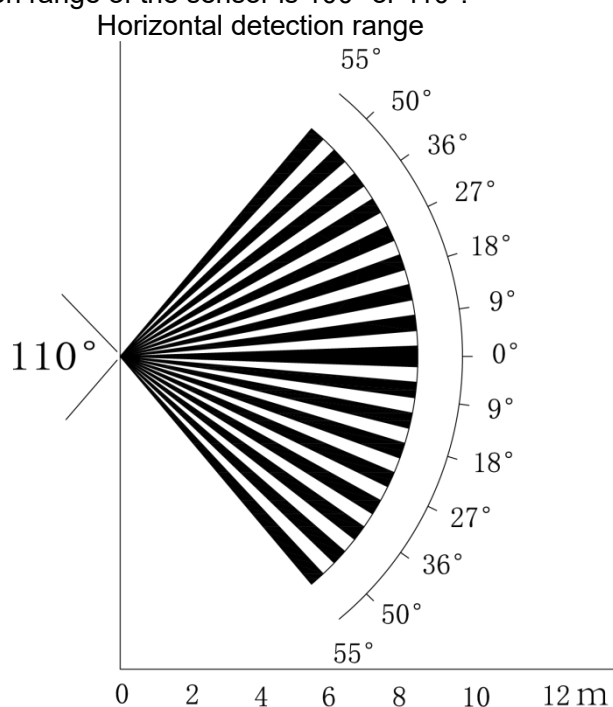


6 Active Deterrence Camera Configuration

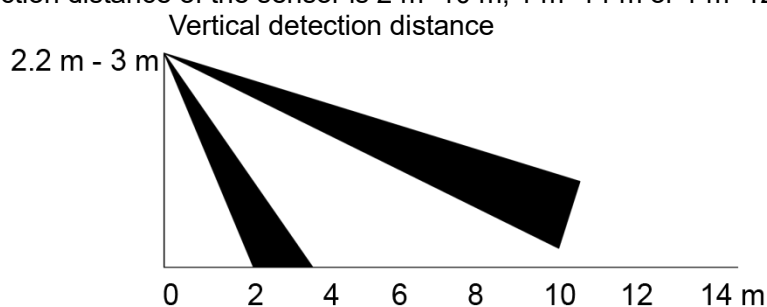
Active deterrence camera can warn off intruders actively with LED even before users are aware of the incidence. Once an intrusion is detected, the LED will be turned on to alert the intruder.

6.1 Detection Range of PIR Detector

The horizontal detection range of the sensor is 100° or 110°.



The vertical detection distance of the sensor is 2 m–10 m, 1 m–14 m or 1 m–12 m.



6.2 Configuring Trigger Mode

On the OSD menu (0), select **Alert > Trigger Mode**.

If you select **Camera Set**, the sound and light alarm will be triggered in the device.

If you select **DVR Set**, the sound and light alarm will be triggered in the DVR.



For more information about **DVR Set**, see user's manual of DVR.

6.3 Configuring Light Warning and Audio Alarm



This function is available only when the trigger mode is set to **Camera Set**.

On the OSD menu (0), select **Alert**. Set **Light Warning** to **ON**, and then enter the item.

- ◇ For **Mode**, You can select from **Lighting** to **Flash**.
If you select **Flash**, you can set the **Flash Frequency** to **Low**, **Medium** or **High**.
- ◇ For **Alert Duration**, you can adjust from 5 seconds to 60 seconds.

On the **Audio Alert** item, set it to **ON**, and then enter the item.

- ◇ In the **Audio** item, you can select from three audios.



You can contact after-sales support to customize alarm audios.

- ◇ In the **Volume** item, you can select from **Low**, **Medium** and **High**.
- ◇ In the **Alert Duration**, you can adjust from 5 seconds to 60 seconds.

6.4 Configuring Smart Dual Illuminators Active Deterrence Camera



For details on configuring smart dual illuminators, see "5 Smart Dual Illuminators Cameras Configuration".

Step Select **Main Menu > Alert**, and then set **Audio**

1Step **Alert** to **ON**.Set the **Volume** and **Alert Duration**.

2



The shortest of alert duration is 5 s, and the longest is 60 s.

Select 5 different audio clips (including **Alarm**, **No parking here**, **Private land no entry**, **Warning zone keep off**, and **Welcome**).

Step

3



You can also DIY the first clip to a .bin file and import to the back-end to use.

Figure 7-3 Main menu

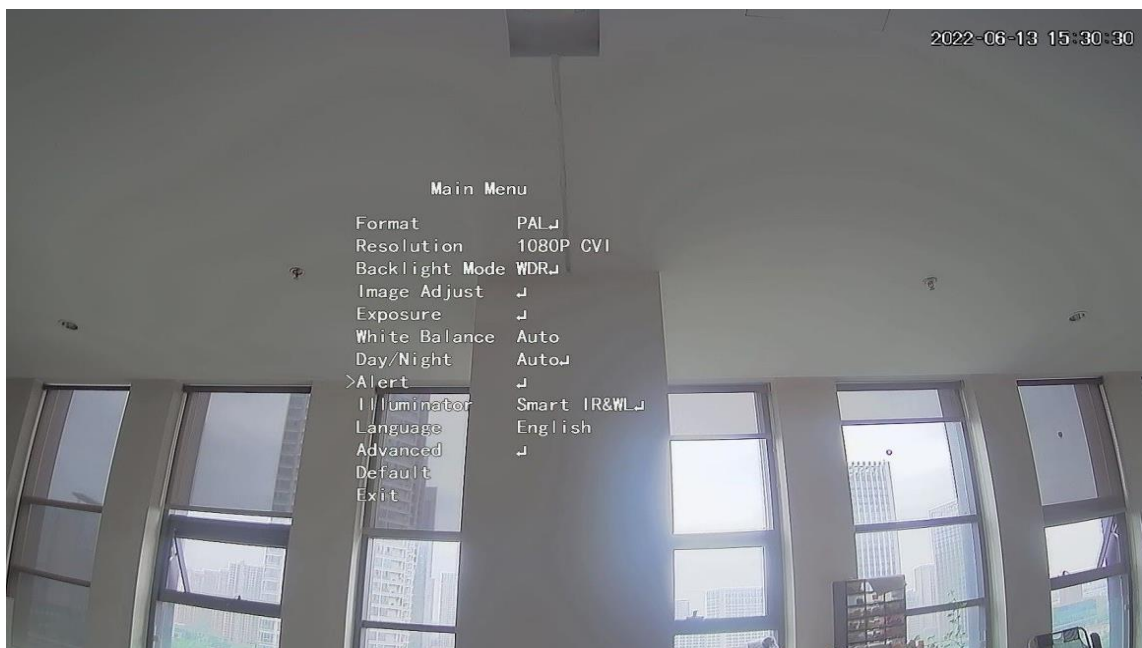


Figure 7-4 Alert

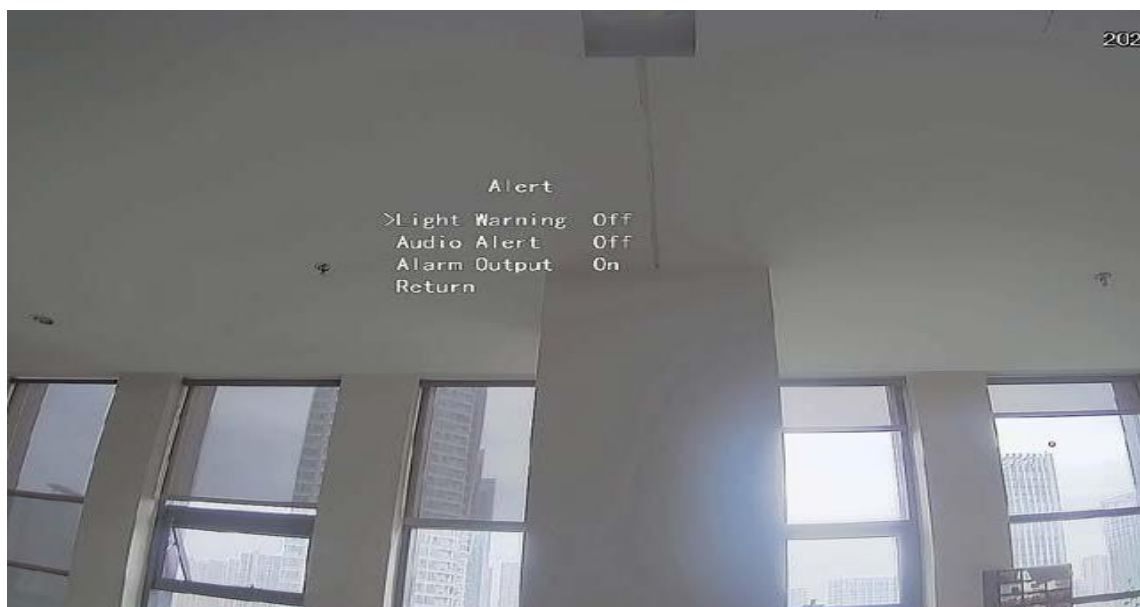


Figure 7-5 Audio alert



Step 4

Select **Main Menu > Alert**, and then set **Light Warning** to **ON**. Set the **Flash Freq** and **Alert Duration**.



The shortest of alert duration is 5 s, and the longest is 60 s.

Figure 7-6 Light warning



7 FAQ

7.1 Long Distance Power Supply

In many scenarios, our clients adopt long distance power supply, transmitting 12V DC to cameras located over 100 m. Such long-distance power supply might cause problems.

Q1: Recurrent restart of devices or even ICR Failure.

Possible reasons: The long power supply cable leads to a large voltage drop on the equipment power supply cable and turning on the IR light at night leads to a further increase of the voltage drop, resulting in restart of the device. After the device is restarted, the ICR is switched to the Day mode by default. By judging the ambient light at night, the device will operate in Night mode, and then the infrared light is turned on, which causes the device to restart again because of undervoltage. Thus, ICR is switched to every 2 seconds, impacting its switching lifespan.

Q2: Unable to restart devices at night, and black screen or restart occurs when switching ICR.

Possible reasons: The long power supply cable leads to a large voltage drop on the equipment power supply cable and turning on the IR light at night leads to a further increase of the voltage drop, resulting in restart of the device and black screen.

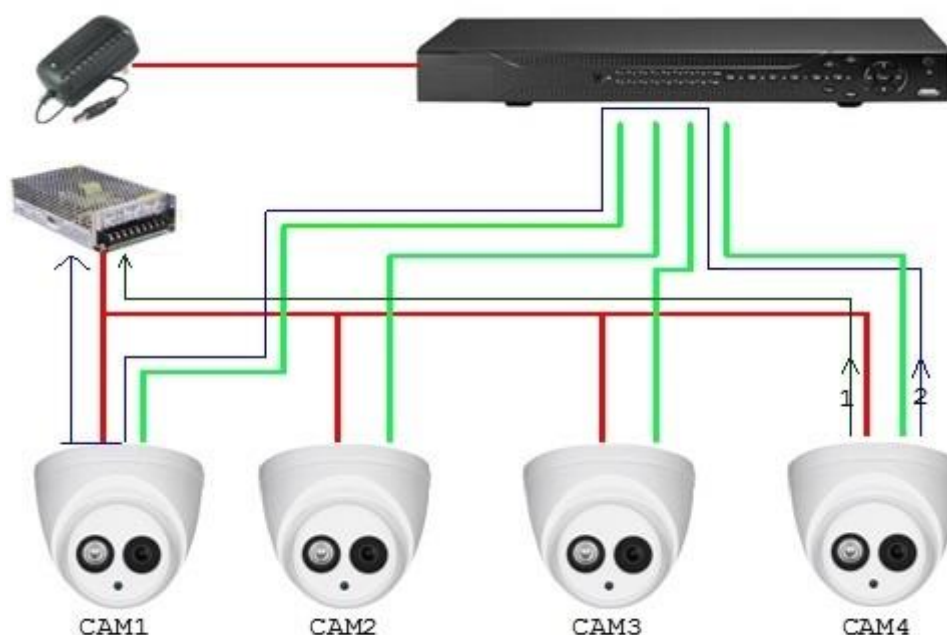
Solution: During construction, when the camera location is far from the power supply, you need to adopt long distance separate power supply or purchase –DP dual power supply to use 24V AC power supply.

7.2 Centralized Power Supply

The typical problem of centralized power supply is that there are obvious black stripes on the device screen, which interferes with the display.

The principle of centralized power supply is as follows:

Principle of centralized power supply



There are two paths for the power output of CAM4, return path 1 and return path 2. Reflow 2 first flows to CAM1, and then flows to the power supply from power supply ground of CAM1. In this way, the reflow of power supply ground CAM4 affects the video ground of CAM1, resulting in interference stripes on the screen. And CAM4 also interferes with CAM2 and CAM3.

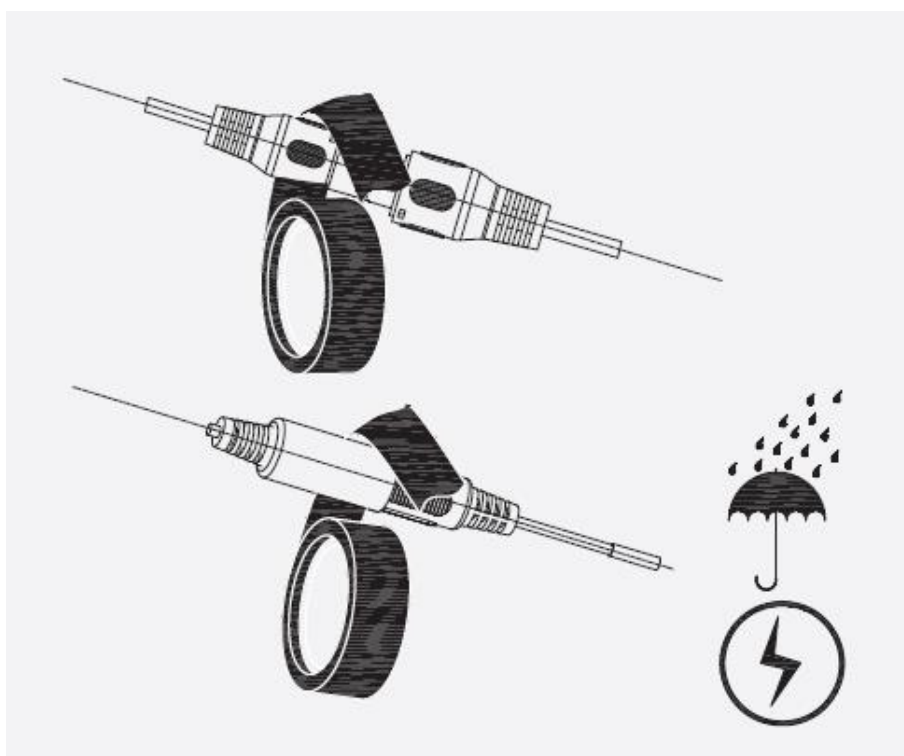
In the same way, CAM1, CAM2 or CAM3 affects other cameras besides itself.

The main reason for the interference of centralized power supply is that the power supply ground of camera is not isolated. To solve this problem: Use dual-power devices with isolation of power supply ground; equip low-power devices with power isolators to block the return path 2 low-power devices can also use power isolators to block the return path 2; use isolated power supplies for each channel, or power the device separately, which are the two recommended methods.

7.3 Connector Waterproof Protection

HDCVI cameras need to be well waterproofed and protected. After installation, wrap the BNC connector and power connector tightly with insulated or waterproofed tape to prevent water and external electromotive forces. When metal casing device is installed on metal surfaces such as elevators and buses, the metal casing should not be in contact with the installation surface to prevent water and external electromotive forces.

Waterproof measures



8 Maintenance



In order to maintain the image quality and proper functioning of the device, please read the following maintenance instructions carefully and hold rigid adherence.

Disassembly and Desiccant Replacement

- Carefully follow the instructions in the manual when performing any disassembly operation about the device; otherwise, it might cause water leakage or poor image quality due to unprofessional disassemble.
- Please contact after-sale service for desiccant replacement if there is condensed fog found on the lens after unpacking or when the desiccant turns green. (Not all models are included with the desiccant).

Maintaining Lens and Lens Protector

- The lens and lens protector are covered with antireflection coating, which could be contaminated or damaged and result in lens scratches or haze images when being touched with dust, grease, fingerprints and other similar substances.
- Do not touch the image sensor (CCD or CMOS) directly. Dust and dirt could be removed with air blower, or you can wipe the lens gently with soft cloth that moistened with alcohol.

Maintaining Device Body

- Device body can be cleaned with soft dry cloth, which can also be used to remove stubborn stains when moistened with mild detergent.
- To avoid possible damage on device body coating which could cause performance decrease, do not use volatile solvent such as alcohol, benzene, diluent and so on to clean the device body, nor can strong, abrasive detergent be used.

Thanks for Choosing CP Plus!



Website: - www.cpplusworld.com

Email id: - sales@cpplusworld.com; support@cpplusworld.com
